

KANE940

Hand-held Combustion Analyser

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1. ANALYSER LAYOUT AND FEATURES

1.1 Instrument Features and Keypad



ON/OFF



MENU

Allows access to all menu functions



PUMP

Turns pump on and off



ENTER

Accepts a command ie enters a menu option



UP

Scrolls up through options ie Fuel



DOWN

Scrolls down through options



STORE

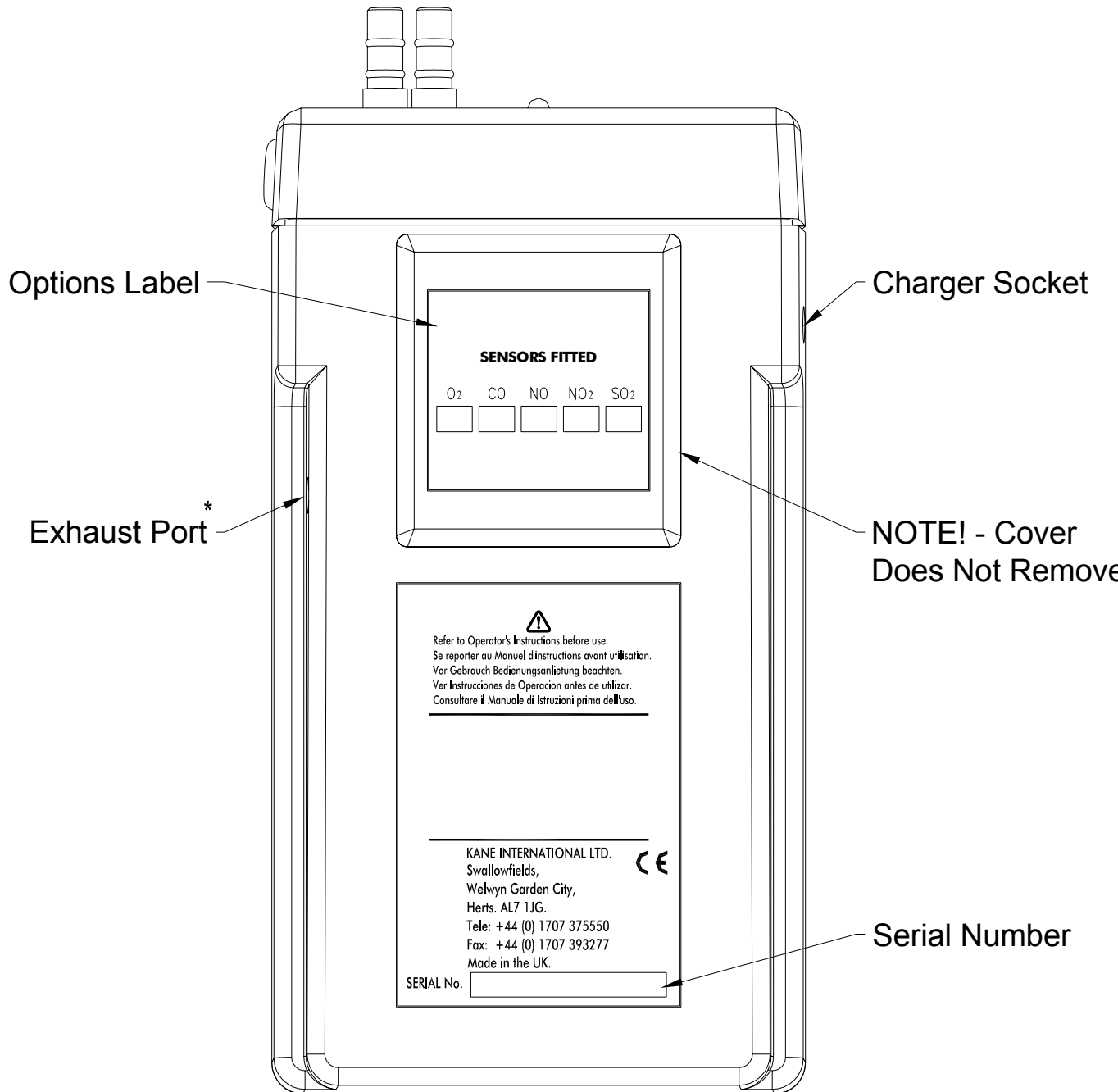
Enters data storage menu



PRINT

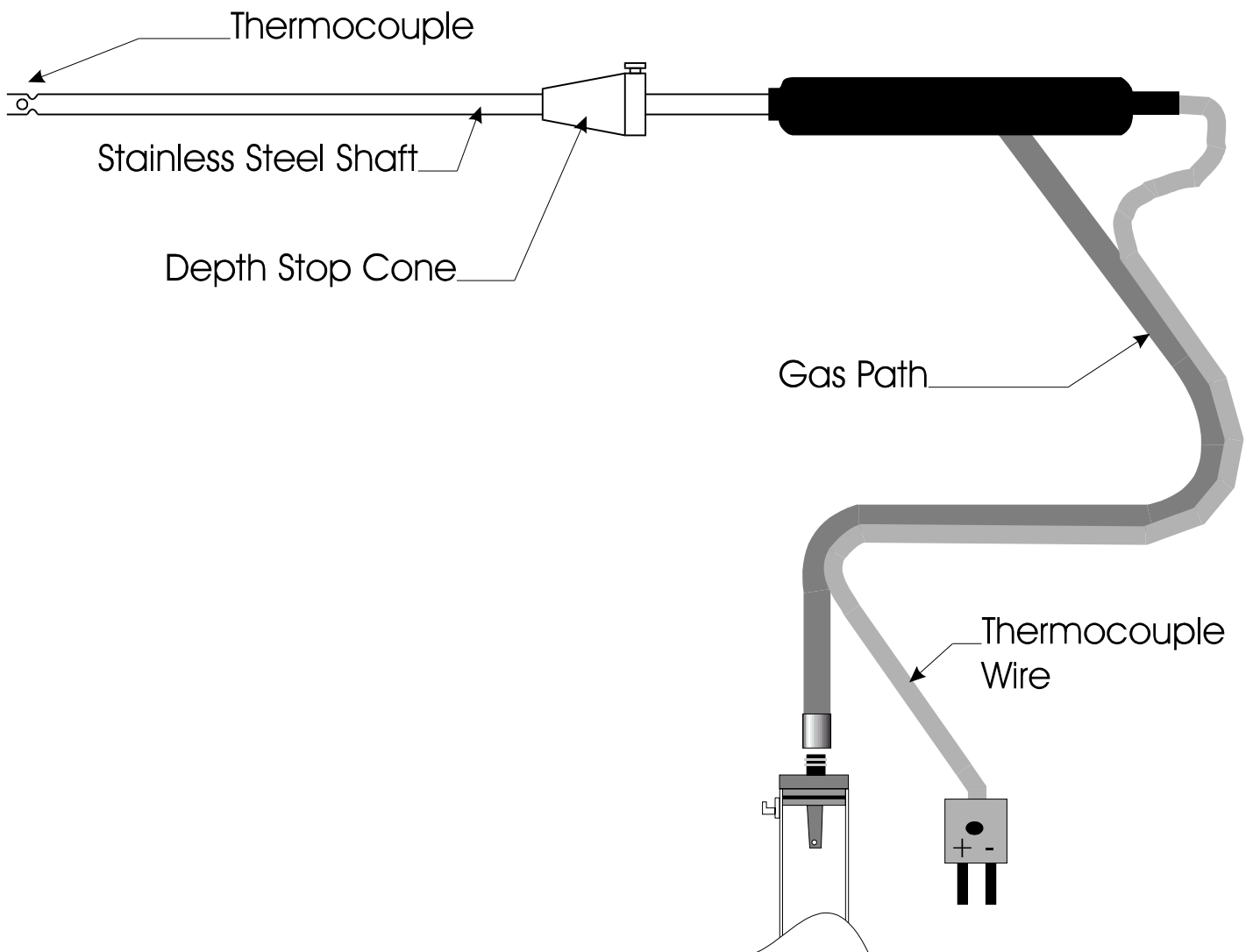
Prints current data

1.2 Instrument Layout (Rear)

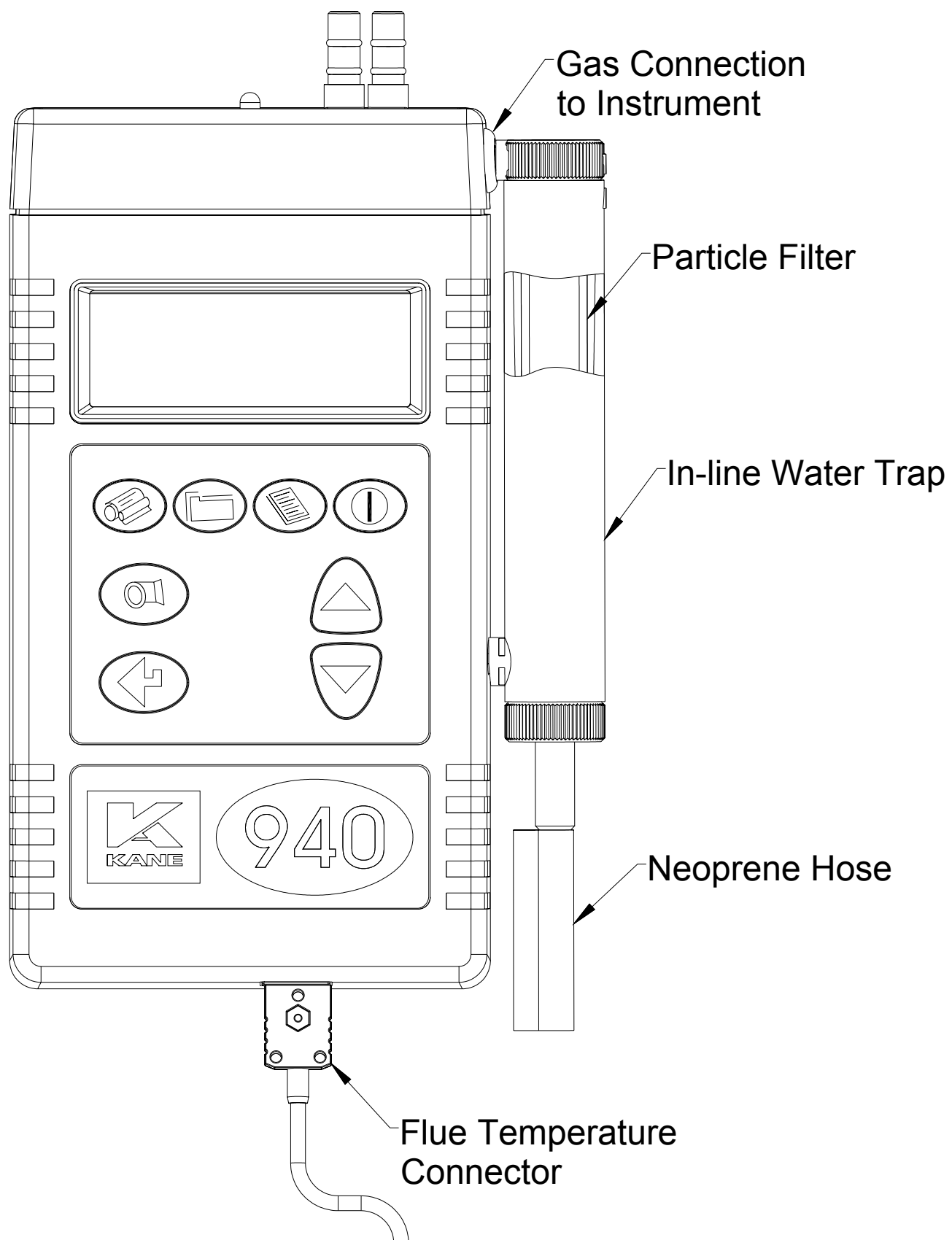


***NOTE! Do not cover exhaust port as this will severely affect analyser operation**

1.3 Standard Probe Configuration



1.4 Analyser Connections





2. SAFETY WARNING

This analyser extracts combustion gases that may be toxic in relatively low concentrations. These gases are exhausted from the side of the instrument. This instrument must only be used in well ventilated locations. It must only be used by trained and competent persons after due consideration of all the potential hazards.

Protection Against Electric Shock (in accordance with EN 61010-1 : 1993)

This instrument is designated as Class III equipment and should only be connected to SELV circuits. The battery charger is designated as:

Class II equipment

Installation category II

Pollution degree 2

Indoor use only

Altitude to 2000m

Ambient temperature 0°C-40°C

Maximum relative humidity 80% for temperatures up to 31°C decreasing linearly to 50%RH at 40°C

Mains supply fluctuations not to exceed 10% of the nominal voltage.

3. FIRST TIME USE

Charge the battery for 12 hours, following this an overnight charge should be sufficient for an average 8 hour day. See Main Parameter displays for Battery Indicator.

The KANE940 has a rechargeable lead acid battery which uses a different charger than other Kane analysers. ***Ensure the correct charger is used or damage may occur to the instrument.***

Check that you have all the items you have ordered.

Take time to read this manual fully.

When using the analyser for the first time you will need to choose from:-

Language selection

Calibration countdown time

CO gas alarm

NOx percentage for calculation

Time and Date

Printed header name and telephone number

The SET UP MENU (Section 5.2.5) gives details of how to change the above settings.

4. NORMAL START UP SEQUENCE

4.1 Every Time You Use The Analyser

BEFORE SWITCH-ON CHECK THAT:

the particle filter is not dirty

the water trap and probe line are empty of water

all hose connections, etc, are properly made

the probe is sampling CLEAN AMBIENT air

the water trap is correctly fitted and the instrument upright

the flue temperature is connected

Switch ON the instrument by pressing 

4.2 Automatic Calibration

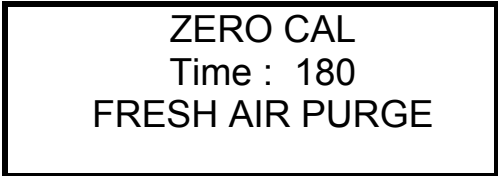
During this sequence the analyser pumps fresh air into the sensors to allow toxic sensors (if fitted) to be set to zero and the Oxygen sensor to be set to 20.9 %.

After switch-on the analyser will briefly display header information :-



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And then show the countdown screen :-



ZERO CAL
Time : 180
FRESH AIR PURGE

The calibration time will count down in seconds to zero. Calibration time may be changed from 2 to 6 minutes. See Set-Up menu section 5.2.5.

Note ! Three minutes is recommended to allow the sensors to stabilise fully. Anything less than this may result in drift of the toxic and oxygen sensors in clean ambient air.

To obtain the quoted specification an instrument should be calibrated with clean ambient air at standard temperature and pressure (STP).

Once the time has reached zero an audible beep will be heard and will show the selected fuel on the following display:-

Press 

NATURAL GAS

PRESS -MENU- KEY

This zeros the toxic sensor and sets Oxygen to 20.9%. The next screen is the **MAIN DISPLAY** of the analyser:-

NETT	C		0.0
O2	%	...	20.9
CO	ppm	...	0000
EFF (G)	%	...	0.0

Use  and  to change the display.

CO2 %		0.0
FLUE C.		0.0
INLT		NOT FITTED
AMBIENT C		21

All parameters are detailed in Appendix A - MAIN DISPLAY PARAMETERS.

4.3 Main Displays

The main display can be changed to show either 4 or 8 parameters at one time. Two options are available when 4 parameters are selected.

- **4 Page Mode** displays 4 lines of data in set format, each page is predefined.
- **Line scroll mode** allows you to customise the display to show the data you require.
- **8 Page Mode** displays 8 parameters on 4 lines in set format, the bottom two can be changed.

Changing between the different modes is detailed in **Display Menu Section 5.2.4**.

4.3.1 4 Page Mode

Use the  and  keys to change the information that is displayed on the screen. The following pages are available.

NATURAL GAS			
DATE	...	07-08-96	
TIME	...	12:31:35	
BATTERY %	..	54	

NETT	C	...	0.0
O2	%	...	20.9
CO	ppm	..	0000
EFF (G)	%	...	0.0

CO2	%	...	0.0
FLUE	C	...	0.0
INLT	NOT FITTED		
AMBIENT	C	...	21

CO/CO2	R	..	0.0001
P INDEX	%	...	0.01
XAIR	%	...	0.0
Prs	mbar	0.00	

This screen only displayed on an analyser fitted with an NO sensor

NO	ppm	..	0000
NOx	ppm	..	0000
NOx calc%	...	5	
O2 ref	%	...	3.0

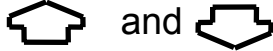
TIP - In 4 page mode only  turns the backlight ON and OFF.

4.3.2 Line Scroll Mode

Line scroll mode allows you to customise the display.

Use the  and  keys to change the bottom line of the display. Once the correct line is displayed press  to confirm and move the line up. Select the next parameter and repeat until all lines display the desired parameters.

Change bottom line using



NETT	C	...	0.0
O2	%	...	20.9
CO	ppm	..	0000
CO2	%	...	0.0

 to select and move
parameter up

O2	%	...	20.9
CO	ppm	..	0000
CO2	%	...	0.0
CO2	%	...	0.0

Select next parameter.
Repeat above until
display reads desired data

O2	%	...	20.9
CO	ppm	..	0000
CO2	%	...	0.0
CO/CO2	R	..	0.0001

4.3.3 8 Page Mode

Displays 8 parameters on the screen at one time. Symbols used in this mode are different to those used in 4 page and line scroll modes and are detailed in Appendix A - MAIN DISPLAY PARAMETERS.

O2	:20.9 %	CO2	: - -
CO	: 0ppm	Eff	: - -
PI	: - -	ΔT	: 0C
λ	: - -	Tf	: 21C

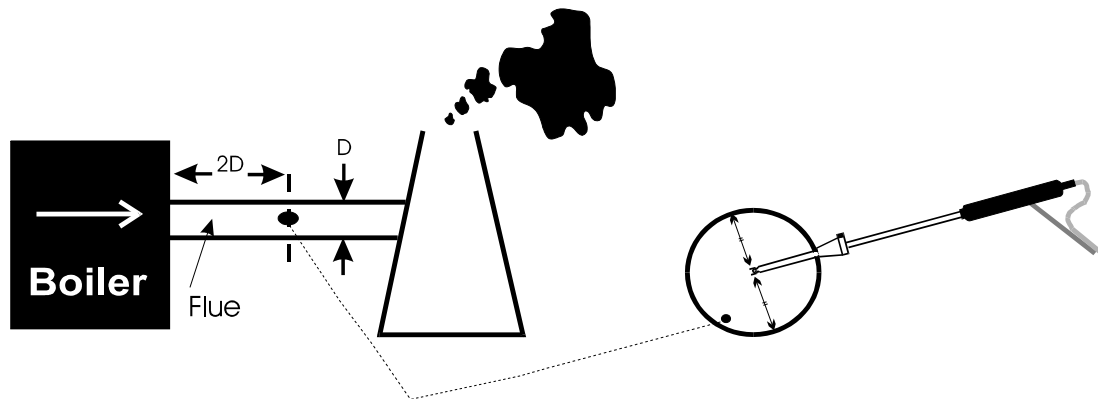
The bottom line of the display can be changed to display other parameters.

Use the  and  keys to change this line.

4.4 Sampling the Flue Gas

Once the automatic calibration procedure has been completed and the specific fuel has been selected (See SELECT menu) the probe can be inserted into the desired sampling point.

It is recommended that the sampling point be located at least two flue diameters downstream of any bend and that the probe tip is in the centre of the flue. With balanced flues and other domestic units the probe should be positioned far enough into the flue so that no air can 'back flush' into the probe. This will be indicated by a low oxygen reading and/or a low 'Poison Index' reading.



The probe depth stop cone provided with the instrument allows the probe to be used in holes whose diameters range from 8 mm to 21 mm ($\frac{5}{16}$ to $\frac{13}{16}$ inch).

The standard probe is rated at 650°C. Temperatures of up to 1100°C (2012°F) can be accommodated using an optional high temperature probe.

TIP To conserve battery power, switch off the pump when you are not taking a measurement. Use the  key to turn ON and OFF the pump.

4.5 Taking a Pressure Reading

With the optional pressure module fitted a flue draught measurement can be made at any time.

Connect the standard probe to the pressure sensor inlet and the probe in the flue.

The pressure reading will be displayed :-

CO/CO2 R ..	0.0001
P INDEX % ...	0.01
XAIR % ...	0.0
Prs mbar	0.00

To perform a combustion test and display draught pressure at the same time a special probe is required. Contact Kane International or Authorised Distributor for details.

4.6 Regular Checks During Sampling

Care must be taken at all times not to exceed the analysers operating specifications, in particular ensure the following :-

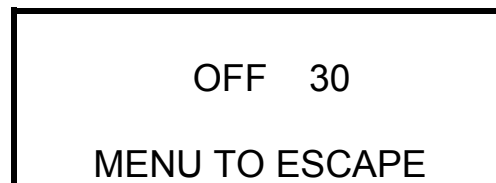
- Do not exceed the maximum temperature of the flue probe.
- The analyser internal temperature does not exceed normal operating range, typically 0-40°C.
- DO NOT PLACE THE INSTRUMENT ON A HOT SURFACE.
- The water trap is vertical at all times. Water condenses in the probe line and can quickly fill the water trap when the probe is moved. Take care and watch the water trap closely.
- The in-line particle filter is clean and does not become blocked.

4.7 Normal Shutdown Sequence

DO THIS EVERY TIME YOU USE THE ANALYSER

Remove the probe from the flue - **TAKE CARE ! THE PROBE WILL BE HOT** - and allow it to cool naturally. Do not immerse the probe in water as this will be drawn into the analyser and damage the pump and sensors.

Once the probe is removed from the flue press  and the analyser will count down from 30 to switch off.



If you have not finished but press  by mistake, you can press  to return to normal operation and not switch OFF.

4.8 Electromagnetic Compatibility

The European Council Directive 89/336/EEC requires that electronic equipment does not generate electromagnetic disturbances that exceed defined levels and has an adequate level of immunity to enable it to be operated as intended. The specific standards applicable to this product are detailed in the appendices.

Since there are many electrical products in use that pre-date this Directive and may emit electromagnetic radiation in excess of the standards defined in the Directive there may be occasions where it would be appropriate to check the analyser prior to use.

The following procedure should be adopted:

Go through the normal start up sequence in the location where the equipment is to be used.

Switch on all localised electrical equipment that might be capable of causing interference.

Check that all readings are as expected. (A level of disturbance in the readings is acceptable). If not adjust the position of the instrument to minimise interference or switch off, if possible, the offending equipment for the duration of the test.

N.B. Maximum cable lengths must be less than 3 metres.

At the time of writing this manual (November 2005) Kane International Ltd is not aware of any field based situation where such interference has ever occurred and this advice is only given to satisfy the requirements of the Directive.

5. MOVING THROUGH THE MENUS

5.1 Basic Operation

From the MAIN DISPLAY

NETT	C		0.0
O2	%	...	20.9
CO	ppm	...	0000
EFF (G)	%	...	0.0

Press  to access the MAIN MENU

MAIN MENU			
1	SELECT	3.	DISPLAY
2.	UNITS	4.	SETUP

Press  and  to move cursor up and down

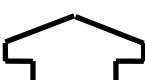
MAIN MENU			
1.	SELECT	3.	DISPLAY
2	UNITS	4.	SETUP

Press  to access selected Menu

MAIN MENU			
1	SELECT	3.	DISPLAY
2.	UNITS	4.	SETUP

Press  to select parameter

F	FUEL	:	LIGHT OIL
	O2 Ref	:	OFF
	SMOKE	:	OFF
	RESET	:	NO

Use  and  to change setting i.e. fuel selected

FUEL	:	N	NATURAL GAS
O2 Ref	:		OFF
SMOKE	:		OFF
RESET	:		NO

Press  to enter value and move to next parameter

FUEL	:		LIGHT OIL
O	O2 Ref	:	OFF
	SMOKE	:	OFF
	RESET	:	NO

Press  to save settings and return to the MAIN MENU

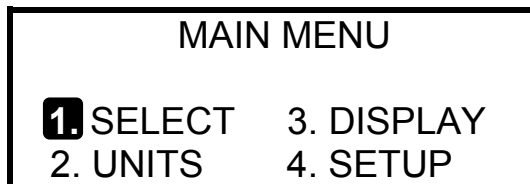
MAIN MENU			
1	SELECT	3.	DISPLAY
2.	UNITS	4.	SETUP

Press  to return to the MAIN DISPLAY

5.2 Menu Options and Settings

5.2.1 Main Menu

The MAIN MENU consists of 4 sub menus which are shown below and detailed on the following pages.

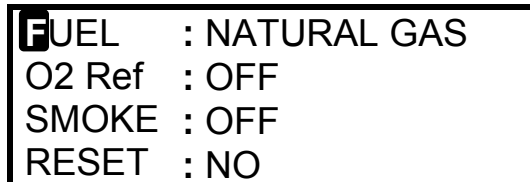


All sub-menus are accessed using  and exited using 

The  and  keys move the cursor within a menu and allow parameters to be changed.

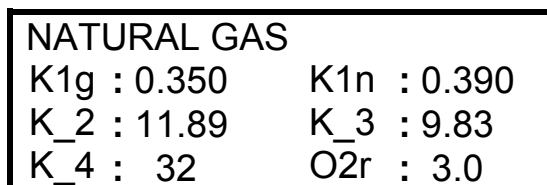
TIP Holding down one of these keys scrolls through the data quicker.

5.2.2 Select Menu



This menu allows selections to be made for the parameters detailed below.

FUEL : Select the fuel being used by the boiler from either a standard fuel stored in the analyser or by entering the user fuel. Once the correct fuel has been selected press  to view the fuel constants.



Calculation of fuel constants are detailed in the Appendix. Fuel constants will have to be calculated before a user fuel can be entered.

To enter the user fuel select 'User Fuel' and Press



USER FUEL			
K1g	:	0.000	K1n : 0.000
K_2	:	0.00	K_3 : 0.00
K_4	:	00	O2r : 00

Use  and  to select the correct value.

USER FUEL			
K1g	:	0.350	K1n : 0.000
K_2	:	0.00	K_3 : 0.00
K_4	:	0	O2r : 00

Use  to move to the next parameter, repeat above until all parameters are correct. Press  to return to SELECT menu.

O2 Ref : Toxic gas measurements can be referenced to defined oxygen levels. Reference values can be set from 1-20%, to AUTO or more normally to the default value - OFF. Setting to AUTO uses the figure in the FUEL constants data.

Oxygen referencing is required by some regulations such as TA-LUFT. If a reference value is selected then toxic gas measurements will be displayed with the symbol (n) attached to the reading. i.e. CO(n)

What does Oxygen reference mean ?

If 3 % O₂ reference is selected and 5 % O₂ is measured in the flue then toxic gas values will be recalculated as if 3 % were measured. The equation for referencing is detailed in the Appendix.

Oxygen referencing prevents false readings being submitted, e.g. allowing more air into the boiler will increase the oxygen level in the flue and hence dilute any toxic gas reading. Oxygen referencing gives readings as if they were undiluted.

SMOKE: Allows the user to enter a smoke test number from 0-9. This value will be printed on the standard printout. Default value is OFF.

RESET : Allows the user set the Oxygen to 20.9% and zero the toxic sensors without turning the analyser off.

Selecting YES and  will display the following screen.

RESET SENSORS
O2 % : 20.9 CO & NO = 0
PRESS ENTER
MENU TO ESCAPE

After pressing  the analyser will count down for 5 seconds and then return to the main display.

WARNING : The sensors must only be reset if you are sure they have been sampling fresh air for at least 3 minutes. Errors in measurement will occur if the sensors are reset during or just after sampling.

5.2.3 Units Menu

 TEMP : C
GAS : ppm
PRESS. : mbar
EFF. : GROSS

Allows all displayed units to be changed.

TEMP : Choose selections from Centigrade °C or Fahrenheit °F.

GAS : Changes the toxic gas measurement units. Select from volumetric readings, parts per million (ppm) or mass flow reading milligrams per cubic meter (mg/m³) .

PRESS. : Flue draught can be displayed in millibar (mbar), hectaPascals (hPa), millimeters water gauge (mmWG) or inches water gauge (in WG).

EFF. : Efficiency can be selected for Gross or Net values. Gross efficiency assumes latent heat of vapourisation is lost in the boiler and hence will be lower than Net efficiency. For Natural Gas the difference will be approximately 11%.

5.2.4 Display Menu

L IGHT	: OFF
MODE	: 8-PAGE
CONTRAST	: DEFAULT

Allows the configuration of the display to be changed.

LIGHT : Choose from ON or OFF.

MODE : Select 4 or 8 Page Mode or Line Scroll Mode as detailed in section 4.3 Main Displays.

CONTRAST : The contrast is set to a DEFAULT value or can be adjusted
↑ LIGHTER or ↓ DARKER. Use the  and  keys to adjust.

5.2.5. Set-Up Menu

The set up menu allows the following parameters to be set / altered.

- Language.
- Automatic calibration time
- CO gas alarm
- NOx percentage for calculation
- Date and time
- Printout Header

L ANG	: ENGLISH	ZERO	: 3
CO ALARM	: 400	NOx%	: 5
CALENDAR		HEADER	

Parameter	Description	Settings
LANG :	Changes the analysers displayed and printed language.	ENGLISH SPANISH DUTCH FRENCH ITALIAN
ZERO :	Allows setting of the Autocalibration time in minutes. Care must be taken when changing this parameter as sensors may drift from zero if too short a time is used. Kane International advise 3 minute countdown.	2-6 minutes
CO ALARM :	Allows an alarm level to be set on for the CO reading. This is set as a default at 1000 ppm.	OFF 0-4000 ppm

Once an alarm has been exceeded the display will flash every two minutes warning the user of an alarm state and display the gas concentration. A similar display will be shown during a RECHARGE BATTERY and PUMP OFF alarms.

CO ALARM

1010 ppm

NO REF: Displayed on the Nitric Oxide unit only. Allows the percentage P in the following calculation to be set. The default value set is 5%. Note the percentage allows for NO₂ in a typical boiler.

$$\text{NO}_x = \text{NO} + \text{P}\% \text{ NO}$$

CALENDAR: Allows the user to change the date and time. (24 hour clock).
The following screen will be shown once the parameter is entered:

	hh : mm : ss
TIME :	13 : 53 : 26
FORMAT :	dd : mm : yy
DATE :	03 : 01 : 96

FORMAT : Changes the date format for display and printing.

dd : mm : yy
yy : mm : dd
mm : dd : yy

To change the time position the cursor on **Time** and press . The cursor will now be to the left of the 13 as shown below:

	hh : mm : ss
TIME :	13 : 53 : 26
FORMAT :	dd : mm : yy
DATE :	03 : 01 : 96

Using  and  scroll through the setting options i.e. 0-23.

Once the correct hour is set press  to move to the next parameter, the cursor will move to the left of minutes (53). Move to each parameter until the correct time is set. Pressing  after setting the seconds will return the cursor to the left of the screen.

Format and Date are set in a similar manner.

Header : Allows two lines of 20 characters to be programmed into the analyser. The header appears on the top of the standard printout. This can be used to print your company name and/or phone number.

Name/Phone <u>K</u> ane International (44)-1707-375550 'LEFT' USE STORE KEY
--

The screen above shows the standard header setting with the cursor now shown underlining the K in Kane. By using  and  any letter or number can be chosen.

Once the correct character is displayed, use  to move right to the next. Move along until all characters spell the desired name or phone number. If you need to go back and change a character use  to move left.

Press  to return to the SET UP menu.

6. PRINTING INFORMATION



Supplied as accessories for the KANE940 are an infra-red thermal printer or a dot matrix serial printer. Read the manual supplied with each printer prior to operation. Connections to the KANE940 are detailed below :

- **Infra-red thermal printer** - this does not require a cable to transmit the data but uses an infra-red (IR) link similar to a TV remote control. The IR emitter is positioned on the top of the KANE940 and the bottom of the printer. Ensure they are pointing at each other and within 300 mm, with no obstructions in the way. Data may be lost if transmission is interrupted. Keep the KANE940 pointing at the printer until the printout has finished.
- **Dot matrix serial printer** - requires the supplied serial cable to transmit data. Connect the cable to the 8 pin DIN socket on the top of the KANE940 and the 25 pin D-connector on the printer.

Data can either be printed from a 'live' test or from stored data. Printing of stored data is detailed in STORING AND RETREIVING DATA.

6.1 Printing a 'Live' Test

During a combustion test the KANE940 will print data on request. With the analyser showing the MAIN DISPLAY press  and current data will be sent to the printer.

The display will show the following until data transmission is complete.

***** Printing *****

6.2 Standard Printout

The standard printout is shown below :-

```
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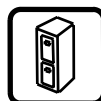
TEST 36
DATE:      01-01-96
TIME:      15:46:52
NATURAL GAS

NET C . . . . . 2
O2 % . . . . . 20.3
CO ppmn 02 > 20%
EFF % (G) . . . . 87.8

CO2 % ° . . . . . 0.3
FLUE C ° . . . . . 24
INTL C ° NOT FITTED
AMBIENT C . . . . 22.6

CO/CO2 R . . . 0.0000
P INDEX % . . . 0.00
XAIR % 02 > 20%
Prs mbar . . . . 0.0
```

7. STORING AND RETRIEVING DATA



The KANE940 can store up to 100 combustion tests. Once stored, the data can be viewed on the display or downloaded to a PC or printer.

7.1 Storing a 'Live' Test

While performing a test and viewing the data on the MAIN display access the STORE menu as follows :-



to access the STORE MENU

```
STORE MENU
MODE      : STORE
LOCATION   :    3
PRESS 'STORE' TO LOG
```

Mode: Select from the following :-

- **STORE** - Allows data to be stored in memory.
- **VIEW / PRINT** - Stored data can be viewed or printed.
- **DELETE** - Clears all data in memory.

Location: Automatically allocates a location in the memory of the instrument for the next test. On the display shown above the next location will be 3.

To store a test set **MODE** to **STORE** and press The current readings will be stored in the analysers memory.

Tip : Make a note of the location number for your particular test as it may be useful when downloading or printing.

7.2 Viewing and Printing a 'Stored' Test

Multiple tests can be printed easily with the KANE940.

Select PRINT under MODE in the STORE menu. This feature is in addition to the VIEW/PRINT, STORE and DELETE options.



to access the STORE MENU

```
STORE MENU
MODE      : PRINT
LOCATION   :  1 TO 10
PRESS 'PRINT'
```

The cursor will move to the first number, use the and to select the location and start printing.

Press to move the cursor to the second number, select the last location to print.

To print the data press . In the screen shown above locations 1 to 10 will be printed.

During printing the following will be shown.

PRINT TESTS
1 to 10
PRINTING TEST 1

NOTE While the display above is shown (i.e. the instrument is printing a test) the keypad is disabled. To exit from printing wait until the current test has finished and the display below is shown:

Press  to exit the print routine. The instrument will return to main display

PRINT TESTS
1 to 10
PLEASE WAIT
MENU TO ESCAPE

7.3 Deleting Data

To delete the data in stored memory press  to obtain the STORE MENU (as above) :-

Press 
to access the STORE MENU

STORE MENU
MODE : ELETE
LOCATION : 3
PRESS 'ENTER' TO DELETE

Press 
to access delete data screen

ENTER to ERASE DATA
MENU to ESCAPE

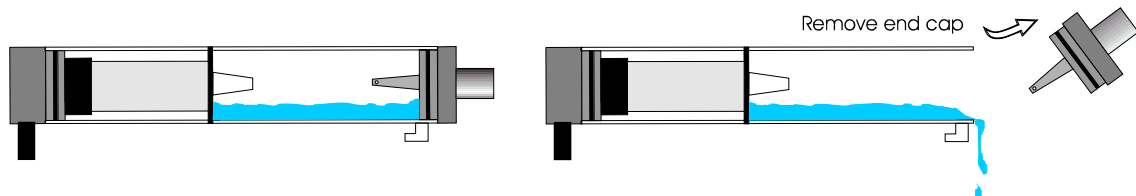
Press  to delete data in memory, press  to exit delete data screen.

8. MAINTENANCE

8.1 Emptying and Cleaning the In-line Water Trap

The in-line water trap should be checked and emptied on a regular basis. Water vapour will condense and gather in the probe line. This may move suddenly to the trap when the probe is moved. Care should be taken at all times.

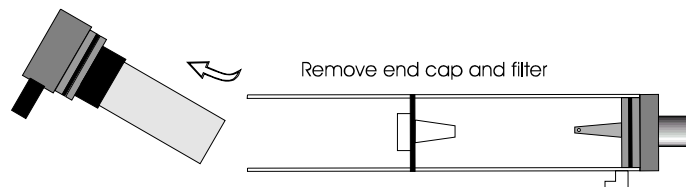
Emptying of the water trap is detailed below :-



Carefully remove the end cap from the in-line housing. Dispose of the condensate in a suitable drain, care must be taken as it could be acidic. If condensate spills onto the skin or clothing, clean off immediately using fresh water, seek medical advice if problems occur.

8.2 Changing the Particle Filter

This is a very important part of the analyser and should be changed regularly. It prevents dust and dirty particles entering the pump and sensors and hence causing damage. The filter **MUST** be changed when it is discoloured.



Remove the end cap from the in-line filter housing. Carefully remove the paper filter element and dispose of it. Clean the inside of the filter housing with a suitable soft cloth. Insert a new filter element onto the spigot in the filter housing and carefully replace the end cap.

9. PROBLEM SOLVING

The following is a list of problems that may occur on the instrument through its operating life. If the cause of the fault is not easy to identify then we advise you contact Kane International Service Department or an International Distributor for expert advice.

Fault symptom	Causes
• Oxygen too high • CO₂ too low	• Air leaking into probe, tubing, water trap, connectors or internal to instrument. • Oxygen cell needs replacing.
• Oxygen Error (FAULT) • Toxic sensor Error (FAULT)	• Calibration time set too short and instrument not allowed to stabilise • Instrument has been stored in a cold environment and is not at normal working temperature. • Oxygen cell or toxic sensors needs replacing.
• Analyser not holding charge • Analyser not charging	• Battery exhausted. • AC charger not giving correct output. • Fuse blown in charger plug.
• Analyser does not respond to flue gas	• Particle filter blocked. • Probe or tubing blocked. • Pump not working or damaged with contaminants. • Probe connected to pressure connector.
• Flue temperature readings erratic	• Temperature plug reversed in socket. • Faulty connection or break in cable or plug.
• Analyser automatically switches off in operation.	• Battery below alarm level. • Ambient temperature above 50°C. • Battery quickly discharging and is faulty.
• Display shows dark lines and no reponse from ON/OFF key.	• Fault has occurred on the instrument electronics and requires resetting. Contact Kane International or Distributor.

10. ANNUAL RE-CALIBRATION

Whilst the sensors have an expected life of more than two years in normal use it is recommended that the analyser is re-calibrated at least annually. This is so that long term drift on the sensors and electronics can be eliminated. Local regulation may require more frequent re-calibration and users should check with appropriate authorities to ensure they comply with relevant guidelines.

11. PRODUCT SPECIFICATION

Parameter	Resolution	Accuracy	Range
Temp Measurement			
Flue Temperature with probe	1.0°C/F	$\pm 2.0^{\circ}\text{C} \pm 0.3\%$ reading	0-600°C 32-1112°F
Inlet Temperature	0.1°C/F	$\pm 1^{\circ}\text{C} \pm 0.3\%$ reading	0-50°C/32-122°F
Pressure	0.01 mbar	$\pm 2\%$ of full scale	+150mbar to -150mbar
Gas Measurement ^{*1}			
Oxygen	0.1%	$\pm 0.2\%$	0-21%
Carbon Monoxide (standard: H compensated)	1ppm	$\pm 20\text{ppm} < 400\text{ppm}$ $\pm 5\% < 5000\text{ppm}$ $\pm 10\% > 5000\text{ppm}$	0-10,000ppm
Carbon Monoxide (high range)	0.01%	$\pm 5\%$ reading from 0.1% to 10%	0 - 10%
Nitric Oxide (standard)	1ppm	$\pm 5\text{ppm} < 100\text{ppm}$ $\pm 5\% > 100\text{ppm}$	0-5000ppm
Nitric Oxide (low range)	1 ppm	$\pm 2\text{ppm} < 30\text{ppm}$ $\pm 5\text{ppm} > 30\text{ppm}$	0 - 100ppm
Nitrogen Dioxide	1 ppm	$\pm 5\text{ppm} < 100\text{ppm}$ $\pm 10\text{ppm} < 500\text{ppm}$ $\pm 5\% > 500\text{ppm}$	0 - 1000ppm
Sulphur Dioxide	1ppm	$\pm 5\text{ppm} < 100\text{ppm}$ $\pm 5\% > 100\text{ppm}$	0-5000ppm
Calculations			
Carbon Dioxide ^{*2}	0.1%	$\pm 0.3\%$ reading	0-99.9%
Losses	0.1%	$\pm 1.0\%$ reading	0-99.9%
Efficiency	0.1%	$\pm 1.0\%$ reading	0-99.9%
Excess Air	0.1%	$\pm 0.2\%$	0-2885.0%
Temp (Nett)	1.0°C/F	$\pm 2^{\circ}\text{C} \pm 0.3\%$ reading	0-600°C/32-1112°F
CO/CO ₂ ratio	0.0001	± 0.0001	0-0.9999
Poison Index	0.01%	± 0.01	0-99.99

Pre-programmed Fuels	Natural gas, Town gas, Gascor, Light Oil, Heavy Oil, Propane, Butane, Anthracite, Coke, Coal, Kinsale Gas
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Dimensions Weight Handset Probe	1kg 220mm x 55mm x 120mm L 420mm x Dia 8mm with stainless steel shaft, type K thermocouple and 3m hose
Ambient operating range	0°C to 45°C/ 10% to 90% RH non condensing
Power supply (battery charger)	Input 110Vac / 220Vac nominal Output: 12Vac off load
Battery Life	>6 hours from full charge

*1 Using dry gases at STP

*2 Calculated

APPENDICES

A - Main Display Parameters

The parameters and their meanings are detailed as follows : -

DATE :	Analyser date. See Set-Up menu section 5.2.5 to change.
TIME :	Analyser time. Use Set-Up menu section 5.2.5 to change.
BATTERY : (BAT)	Displays the battery level from 0-100%. The analyser will flash RECHARGE BATTERY at less than 10 % of charge. With the charger connected the display shows AC ON .
NETT : (ΔT)	Nett temperature calculated by deducting the internal AMBIENT temperature from the measured FLUE temperature. Displays in either °C (C) or °F (F) and will display NOT FITTED (N/F) if flue probe is not connected. If an external INLET probe is used then INLET is deducted from FLUE.
O2 :	Oxygen reading in percentage %.
CO :	Carbon Monoxide reading indicated in ppm or mg/m ³ . If the figures are referenced to oxygen then the display will show CO(n) . See SELECT menu 5.2.2 for oxygen reference. The display will read 'O2 > 20%' if referenced values selected and instrument is in clean ambient air.
EFF (G) :	Combustion Efficiency calculation displayed in percentage. Gross G or Net N can be set see SELECT menu 5.2.3. The calculation is determined by fuel type see Appendix B for calculation. The efficiency is displayed during a combustion test, '- -' is displayed while in fresh air.
CO2 :	Carbon Dioxide calculation determined by the type of fuel. This only shows a reading when a combustion test is being carried out. '- -' is displayed while in fresh air.
FLUE : (Tf) NOT FITTED (N/F) or FAULT (FLT)	Temperature measured by flue gas probe in Centigrade or Fahrenheit. Will show ambient temperature after fresh air calibration and NOT FITTED (N/F) or FAULT (FLT) if probe disconnected.
INLET :	Temperature measured by the optional inlet air probe. This probe is (Ti) plugged into the instrument through the RS232 socket. This figure is used to calculate the NET temperature instead of AMBIENT when fitted.
AMBIENT :	Temperature measured by the internal sensor, used in the NET

(Ta)	temperature calculation if an INLET probe is not fitted.
CO/CO₂ R :	The CO/CO₂ ratio, is the ratio of measured CO divided by calculated CO₂. It gives an indication of the following :- • How good a gas sample the instrument is reading. • How clean the boiler is running. For example : A new or clean domestic boiler will display a ratio of less than 0.004, a unit in need of cleaning 0.004-0.008 and a unit in need of major overhaul will show greater than 0.008. This only shows a reading when a combustion test is being carried out. ‘- -’ is displayed while in clean ambient air.
P INDEX :	The CO/CO₂ ratio expressed as a percentage %, called the ‘Poison (PI) Index’ i.e. P INDEX % = 100 x CO/CO₂. ‘- -’ is displayed while in clean ambient air.
XAIR % : (λ)	Excess air calculated from the measured oxygen and type of fuel used. During a combustion test ‘O₂ > 20%’ will be displayed while in clean ambient air.
Prs :	Flue draught pressure reading. Displayed when pressure sensor fitted. See UNITS menu 5.2.3. for scales.
NO:	Nitric Oxide reading in ppm or mg/m³. Displayed when Nitric Oxide sensor fitted. Also displayed as NO (n) when referenced to oxygen. The display will read ‘O₂ > 20%’ if referenced values selected and instrument is in clean ambient air.
NO_x :	Calculated total Nitric oxides displayed in ppm or mg/m³. Where NO_x = NO + P%NO, note P can be set from 0-9%, default = 5%. See SELECT menu 5.2.2. Also displayed as NO_x (n) referenced to oxygen. The display will read ‘O₂ > 20%’ if referenced values are selected and instrument is sampling clean ambient air.
SO₂ :	Sulphur Dioxide reading in ppm or mg/m³. Displayed when Sulphur Dioxide sensor fitted. Also displayed as SO₂ (n) referenced to oxygen. The display will read ‘O₂ > 20%’ if referenced values selected and instrument is in clean ambient air.
O₂ ref % : (O₂r)	Toxic gas measurements can be referenced to defined oxygen levels. See SELECT menu 5.2.2 for details.

B. COMBUSTION EFFICIENCY CALCULATION

The efficiency calculation is based upon British Standard BS845.

This identifies three sources of loss associated with fuel burning:

Losses due to flue gasses: Dry Flue gas loss,
Moisture and hydrogen
Sensible heat of water vapour
Unburned gas

Losses due to refuse: Combustible in ash
Combustible in riddlings
Combustible in dust

Other losses: radiation
convection
conduction
other unmeasured losses

Net efficiency calculations assume that the energy contained in the water vapour (formed as a product of combustion and from wet fuel) is recovered and the wet loss term is zero. Gross efficiency calculations assume that the energy contained in the water vapour is not recovered.

Since the fuel air mixture is never consistent there is the possibility of unburned/partially unburned fuel passing through the flue. This is represented by the unburned carbon loss.

Losses due to combustible matter in ashes, riddlings, dust and grit, radiation, convection and conduction are not included.

Efficiency Calculation:

Known Data - Fuel: Q_{gr} = Gross Calorific Value (kJ/kg)
 Q_{net} = Net Calorific Value (kJ/kg)
 $K1$ = Constant based on Gross or Net Calorific Value:
 $K1g = (255 \times \% \text{Carbon in fuel}) / Q_{gr}$
 $K1n = (255 \times \% \text{Carbon in fuel}) / Q_{net}$
 $K2$ = % max theoretical CO_2 (dry basis)
 $K3$ = % Wet Loss
 H_2 = % Hydrogen
 H_2O = % Water

Measured Data: T_f = Flue Temperature
 T_i = Inlet Temperature
 O_{2m} = % Oxygen in flue gas
 O_{2r} = Oxygen reference %

Calculated data:

- Tnet = Net Temperature
- % CO₂ content in flue gas
- % Dry Flue Gas losses
- % Wet losses
- % Unburned carbon loss
- % Efficiency

Tnet = Flue Temperature - Inlet Temperature

Dry flue gas loss % = $20.9 \times K1 \times (T_{net}) / K2 \times (20.9 - O_2m)$

Wet loss % = $9 \times H_2 + H_2O / Q_{gr} \times [2488 + 2.1T_f - 4.2 T_i]$

simplified = $[(9 \times H_2 + H_2O) / Q_{gr}] \times 2425 \times [1 + 0.001 T_{net}]$

Wet loss % = $K3(1+0.001 \times T_{net})$

Where K3 = $[(9 \times H_2 + H_2O) / Q_{gr}] \times 2425$

Net Efficiency % = 100 - dry flue gas losses
= $100 - 20.9 \times K1n \times (T_{net}) / K2 \times (20.9 - O_2m)$

Gross Efficiency % = 100 - {dry flue gas losses + wet losses}
= $100 - \{[20.9 \times K1g \times (T_{net}) / K2 \times (20.9 - O_2m)] + [K3 \times (1 + 0.001 \times T_{net})]\}$

Excess Air = $[(20.9\% / (20.9\% - O_2m\%)) - 1] \times 100\%$

CO₂% = $[(20.9 - O_2m) \times K2 / 20.9]$

Unburned fuel Loss % = $K4 \times CO / (CO + CO_2)$ Note: CO scaled in %

Where K4

- = 70 for coke
- = 65 for anthracite
- = 63 for Bituminous coal
- = 62 for coal tar fuel
- = 48 for liquid petroleum fuel
- = 32 for natural gas

The formula for K4 is based on the gross calorific value Q_{gr}. To obtain the loss based on net calorific value multiply by Q_{gr}/Q_{net}. Since this loss is usually small this conversion has been ignored. This loss is subtracted from the efficiency.

Oxygen Reference $CO(n) = CO \times \frac{(20.9 - O_{2r})}{(20.9 - O_2m)}$

C. CALCULATION OF FUEL DATA

For any fuel not specified by Kane International the net calorific value, gross calorific value and composition should be obtained from the fuel supplier.

The following fuel data has been calculated with reference to the efficiency calculation.

Example 1:

Chemical composition:	C	25%	
	H ₂	3%	
	H ₂ O	50%	
	Q _{net}	8.35 MJ/kg	
	Q _g	9.3 MJ/kg	*
	Max CO ₂	20.4%	

$$\begin{aligned}\mathbf{K1n} &= (255 \times \% \text{ carbon in fuel}) / Q_{\text{net}} \text{ (kJ/Kg)} \\ &= (255 \times 25) / 8350 = \mathbf{0.763}\end{aligned}$$

$$\begin{aligned}\mathbf{K1g} &= (255 \times \% \text{ carbon in fuel}) / Q_g \text{ (kJ/Kg)} \\ &= (255 \times 25) / 9300 = \mathbf{0.685}\end{aligned}$$

$$\mathbf{K2} = \text{Max \% CO}_2 = \mathbf{20.40}$$

$$\begin{aligned}\mathbf{K3} = \text{Wet Loss} &= [(9 \times \% \text{H}_2 + \% \text{H}_2\text{O}) / 9300] \times 2425 \\ &= [(9 \times 3 + 50) / 9300] \times 2425 \\ &= (77 / 9300) \times 2425 = \mathbf{20.08}\end{aligned}$$

$$\mathbf{K4} = \mathbf{65} \text{ (an approximation for wood) } *$$

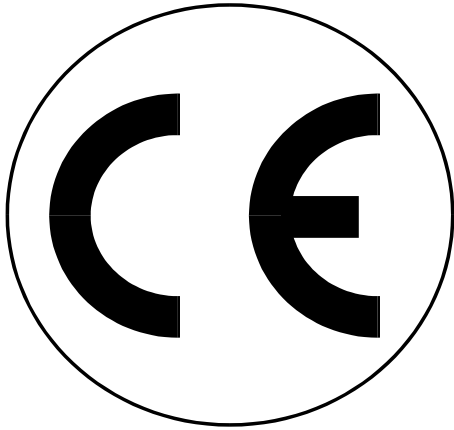
The fuel values to program into the Analyser are as follows:

NATURAL GAS

K1g : 0.763	K1n : 0.685
K_2 : 20.4	K_3 : 20.08
K_4 : 65	O2r : 8.0

* Assumed values in the absence of supplied data. See previous appendix for other fuels.

D. ELECTROMAGNETIC COMPATABILITY (CE) STATEMENT



This product has been tested for compliance with the following generic standards:

EN 61000-6-3

EN 61000-6-1

and is certified to be compliant

Specification

EC/EMC/KI/KANE940 details the specific test configuration, performance and conditions of use.

Please Note: Batteries used in this instrument should be disposed of in accordance with current legislation and local guidelines.